

Shyam Premkumar Shrivastava

A versatile professional with 18 years of experience in R&D, Conceptualization & Development of Microprocessor Products/Solutions, offering excellence in embedded systems, control systems, robotics, and mobile software development

shyam.nexus@gmail.com +91 9158212310 **LOCATION PREFERENCE:** Pune

CORE COMPETENCIES

Technical Project & Program Execution
Embedded Systems Architecture
Cross-Functional Engineering Leadership
Risk-Based Planning & DFMEA
Technology Roadmap & Platform Strategy
New Product Development (NPD Lifecycle)
Work Breakdown Structure (WBS) & Milestone Governance
Client & Stakeholder Engagement
Requirements Engineering & System Decomposition
Agile & Structured Development Models
Escalation Management & Technical Decision Making

WEALTH OF EXPERTISE

Embedded & System Architecture

- SoC / MCU Firmware Development
- Bare-Metal & RTOS-Based Systems
- Embedded Linux (Yocto, Buildroot)
- Bootloaders & Secure Boot Architecture
- Device Driver Development
- Hardware Schematic Review & PCB Design Awareness

Connectivity & Communication

- RFID / NFC
- GPRS, BLE, GPS, Wi-Fi
- TCP/IP, HTTP, MQTT, UDP,
- UART, I2C, SPI, RS485, Ethernet
- Modbus, Zigbee, DigiMesh

UI & HMI Systems

- Custom LCDs (Segment, Graphical, OLED)
- EMWIN & GUI Builder

Memory & Hardware Interfaces

- SD Card, NOR Flash, NAND Flash, SDRAM,DDRAM
- Stepper & Servo Motor Control

Sensors

- Image Sensors: On Semi , Galaxy Core , Smart Sense
- ECG , SPO2 , Skin Temperature
- Load Cells , Strain-Guage , MEMS , Accelerometer , Thermocouples

EDUCATION

2005: Bachelor of Engineering (Electronics & Telecommunication) from RITEE, Raipur CG

PROFILE SUMMARY

Technical Project Lead with 18+ years of experience delivering embedded Linux, safety-critical, and security-focused systems across surveillance, fire protection, and medical domains. Currently leading a 12-member cross-functional engineering team at Honeywell, driving architecture, development, certification, and delivery of IP camera platforms, cost-optimized fire panels, and WiFi-enabled wearable health devices.

Expertise includes Embedded Linux (Yocto, Buildroot), secure boot architecture, TPM integration, H.264/H.265 video pipelines, ONVIF compliance, DFMEA-driven risk mitigation, and cybersecurity governance. Recognized as a decisive technical authority for architecture trade-offs and system-level decisions, with strong focus on delivery predictability, process maturity, and scalable platform design.

Actively seeking to transition into a formal Technical Engineering Manager role where I can scale engineering capability while continuing to contribute at the system architecture level.

TECHNICAL SKILLS

- Jira
- Confluence
- Accolade
- Microsoft Project Planner(MPP)
- Threat Modeling
- UML Diagrams
- MPLABX
- STM32CUBEMX
- Arduino
- AVR Studio
- Github Enterprise
- Gitlab
- Sonarqube
- Black Duck Scan
- Keil
- IAR Embedded Workbench
- VS- Code
- Platform IO
- Kleopatra
- ESP-IDE

CAREER TIMELINE

Since Feb 2024 — **Honeywell (HIIPL) as Project Lead**

2016–2024 — **SAI Agrotel, Pune as Manager Technical**

2012-2016 — **Precise Technology, Pune as Senior Design Engineer**

2008-2012 — **NESD Labs/ Nexus Embedded, Pune as Senior R&D Engineer**

2007-2008 — **Transworld- Compressor Technologies, Pune as R&D Engineer**

WORK EXPERIENCE

Since Feb'24: Honeywell, Pune as Project Lead – Fire, Security and Access

- Lead 12-member cross-functional engineering team delivering safety-critical and surveillance platforms.
- Define scalable system architecture (Hardware + Firmware + Linux BSP) using Yocto and Buildroot enabling platform reuse.
- Drive H.264/H.265 video integration and ONVIF interoperability compliance for enterprise IP camera platforms.
- Architect and implement secure boot chain (ROM → Bootloader → Kernel → RootFS) with TPM-based device hardening.
- Lead STQC and BIS certification lifecycle with zero critical audit rejections.
- Conduct DFMEA and structured risk mitigation improving early-stage defect detection by ~25%.
- Contributed to cost-optimized fire panel architecture achieving ~10–15% BOM reduction.
- Designed WiFi-enabled rechargeable wearable monitoring device with ~20% improved power efficiency.
- Reduced vulnerability remediation cycle time by ~35% through structured cybersecurity workflows.
- Drive milestone governance achieving >95% schedule adherence across programs.
- Own performance evaluations for 6 engineers and mentor subsystem leads.

Apr'16 JAN – JAN'24 : SAI Agrotel, Pune as Manager Technical

Key Result Areas:

- Gathering & understanding client requirements of customer and devising project plans with proper timeline and deliverables
- Training and mentoring multiple teams to acquire new skills required for successful and timely completion of projects
- Working with a team of Firmware Developers and Hardware Designers to complete development of devices to be used for Plant Automation in Rock Quarry and Mining Industry.
- Worked very closely with customers from various regions in the world and established strong customer relationships
- Collaborated with internal Field Application Engineers (FAE) and Sales Teams to resolve product sales issues and address urgent technical issues reported by customers
- Steered the complete application development cycle from creation of Statement of Work (SOW) to post-delivery support of the product/project
- Anticipated and developed a number of product ideas which became market relevant 12 months down the line
- Delivered many Product Services Projects where new features based on customer requirements were added to base layer products
- Put strong emphasis on quality and stability improvements to the product; led a dedicated team of engineers whose main focus was on improving product stability
- Ensured systematic tracking of issues through a Bug Tracking System
- Developed system specifications based on customer interview data, competitor datasheet benchmarking, competitor benchmark testing, and inputs from internal experts
- Formulated a project schedule based on time estimates of Work Breakdown Structure (WBS) tasks; maintained & revised the Gantt chart as required over the course of the project
- Participated in monthly technical reviews on project status and technical issues with Engineering Directors and the VP of Technology
- Presented quarterly program progress reviews to executive leadership; reports included status on budget and schedule variance as well as compliance to requirements
- Allocated merit increases within the annual merit budget of the group based on individual performance
- Directed the development of the technology roadmap for the division by identifying desired future products, technology and Internet Protocol (IP) required to develop those products
- Administered business planning & analysis for assessment of revenue potential, identified target prospectus, audiences in the prospective markets and implemented marketing strategies to achieve targeted business volumes
- Led a global team of 7 software, electrical, and mechanical engineers through the corporate stage-gate new product development process from conceptualization to product release, utilizing methods prescribed by Design for Six Sigma (DFSS) Developed global accounts and engagements in the Industrial Engineering, Consumer Goods and High Tech sectors

PREVIOUS EXPERIENCE

Aug'12-Apr'16: Precise Technology, Pune as Senior Design Engineer

May'08- Jul'12: NESD Labs/ Nexus Embedded, Pune as R&D Engineer

Mar'07-Apr'08: Transworld- Compressor Technologies, Pune as R&D Engineer

PERSONAL DETAILS

Date of Birth: 10th August 1983
Languages Known: English, Hindi
Current Address: Shyam ShrivastavaB-304 Dreams Nandini Society,Shewalewadi, Pune 412307
Email : shyam.nexus@gmail.com
Website : www.shyamshrivastava.com
Linkedin: https://www.linkedin.com/in/shyamshrivastava/

Kindly refer Annexure for Project Details

ANNEXURE

Project : Indigenous IP Camera Platform – Enterprise Surveillance
Feb 2025 – March 26

- Led development of in-house Linux-based IP Camera platform for regulated deployments
- Architected secure boot and TPM-based device hardening
- Integrated H.264/H.265 encoding and ONVIF interoperability
- Customized Linux BSP (Yocto/Buildroot) and driver stack
- Implemented Edge AI analytics and secure firmware upgrade mechanism
- Drove STQC & BIS certification with zero critical rejections
- Conducted DFMEA-based risk mitigation improving early defect detection
- Reduced escalation dependency via structured subsystem ownership
- Novatek SoC , On-Semi Image Sensor, Macronix Nand Flash

Project : VECE Activity – Fire Panel (Cost Optimization Initiative)
Feb 2024 - Sep 2024

- Analyzed hardware and firmware architecture to identify cost-reduction opportunities without compromising safety compliance
- Conducted component-level BOM optimization achieving ~10–15% cost improvement
- Performed DFMEA-driven risk analysis to ensure safety-critical integrity was preserved
- Reviewed schematic and firmware interactions to remove redundant design blocks
- Balanced cost optimization with certification constraints and reliability requirements
- Supported validation planning to ensure compliance with regulatory fire safety standards
- STM32 MCU , Xpedition Suite by Mentor Graphics

Project: Wearable Smart Patch – WiFi-Enabled Vitals Monitoring Device
Sep 2024 - till date

- Architected MCU-based firmware integrating ECG, SpO2, heart rate, and temperature sensors
- Designed secure WiFi data synchronization framework for cloud-based monitoring
- Implemented power-optimized firmware improving battery life and recharge cycles
- Integrated wireless communication stack (TCP/IP, MQTT) for reliable remote monitoring
- Ensured secure data transmission for sensitive health information
- Conducted system-level risk analysis and structured validation planning
- Led cross-functional coordination across hardware, firmware, and validation teams
- Nordic Semiconductor, Espressif ESP32-S3 , ESP-IDF , Analog Devices Bio Sensor Hub

STQC & BIS Certification – Security Camera Portfolio
July 2025 – Till Date

- Coordinated end-to-end STQC and BIS certification process for IP camera platforms
- Ensured cybersecurity compliance including secure boot validation and vulnerability remediation
- Prepared structured documentation for regulatory audits and lab evaluations
- Collaborated with QA and lab partners to resolve compliance observations
- Implemented internal certification-readiness framework reducing audit preparation effort
- Aligned firmware security posture with government cybersecurity guidelines
- SoC : Novatek . Innofusion , Ambrella ,
- Image Sensor: Galaxy core , Smartsense , On-Semi

Some other Notable Projects:

Project: Baby Scale
Cardinal Detecto (USA)
May 2023 to Dec 2023
Temperature compensated Weighing Machine.
Low weight range high precision machine.
Welch Allyn USB Interface for Medical Devices.

Project: 825 Programmable Scale
Cardinal Detecto (USA)
Mar 2023 to Dec 2023
iMx.8 Fedora Yocto 10 Inch LCD Multiple Scale configuration PHP 8 , Apache
Embedded C++, Wayland Architecture.

Project: Med Vue
Cardinal Detecto (USA)
DETECTO's MedVue® medical weight analyzer is utilized on select clinical scales, including DETECTO?s wheelchair, bariatric, and stretcher scales. The MedVue® features a vivid 3.5-in-wide by 3-in-high backlit LCD display screen with nearly 1-in-high weight digit readouts for easy visibility. The MedVue's 350-degree swivel mounting bracket allows the display to be tilted and rotated into position wherever it is the most convenient for the medical professional and/or patient to view. A host of other benefits include units locking into kilograms or pounds, 14-digit patient ID which may be entered via the alphanumeric keypad, 99 stored tare weights, 200 stored transactions, 1LOOK display, four different languages available in set-up (English, Spanish, French, and German), and Wi-Fi/Bluetooth models are available with up to 150 ft. wireless range. The MedVue® offers a host of both wired and optional wireless connectivity for sending measurement data to multiple devices all at once.

Project: 195/225 -Navigator
Cardinal Detecto (USA)
Jul 2022 to Jan 2024
STM32 MCU , ESP32 Bootloader , FOTA, Graphical LCD With an optional dual scale input board, up to 3 scales with total weights can be viewed simultaneously. Standard features include IP66 washdown stainless steel enclosure, 4 bi-directional RS232 serial ports, USB-B port, QWERTY keypad, 4 soft keys, 13 function keys, navigation keys, 16 Preset Weight Comparators (PWCs), 8 input control lines, 16 output control lines, piece count, batching, electronic tally roll, and 200 ID database for truck storage. NTEP legal-for-trade and OIML certified. Supports up to 14 350-ohm load cells. C++ , FreeRTOS

Project: Medical Cart
Cardinal Detecto scales USA
May 2021 to Jan 2024
ATSAM7 main processor , ESP32 Co processor. Journaling File System. Touch Screen, LCD , Microchip Harmony V3. Synchronizing Data and Audit Trails/records over a remote server. Alarms for unauthorized Access. Battery Critical Operations. Deep Sleep and multiple Wakeup sources. FOTA Bootloader

Project: Mosaic Controller
Haven Lightings USA
Feb 2020 to Apr 2021
STM32 ESP32 based Light(Led based)Controller. Offline +Online mode. Event Driven , Secure communication , Azure IOT Hub Embedded Web Server

Project: Haven Light Controls (2020)
Haven Lightings USA
Built around STM32 and ESP32 MCU's , this device is designed to control ambience and Scenes using high quality LED's over a Secure Azure Network

Project: Plant Connect Module (2018)
The device is used to provide a superior controller Interface to the earlier developed Integrator module; The device connects to a server and uploads necessary data required by polling from the Integrator devices
Involved: USB2.0 USB MSD, ETHERNET, 4G modem, SNTP, MQTT Secure, SEGGER EMWIN GUI TOOLS KEIL 5, CMSIS RTOS, Controller: LPC1788

Project : Beltscale Integrator (2017)
The device used to monitor weight /rate control of materials transferred over Conveyor belts in Mining Areas. It also provides the facility of blending control when devices are connected in Master-Slave configuration
Involved: Modbus, Ethernet, SNTP, and HTTP. TCP, SPI UART, NOR FLASH, USB2.0 MSD, EMWIN GUI BUILDER , RTX RTOS, ADC Data acquisition, PID control system , PLC Interface